Worksheet 1: Lipid Digestion — Concepts and Applications

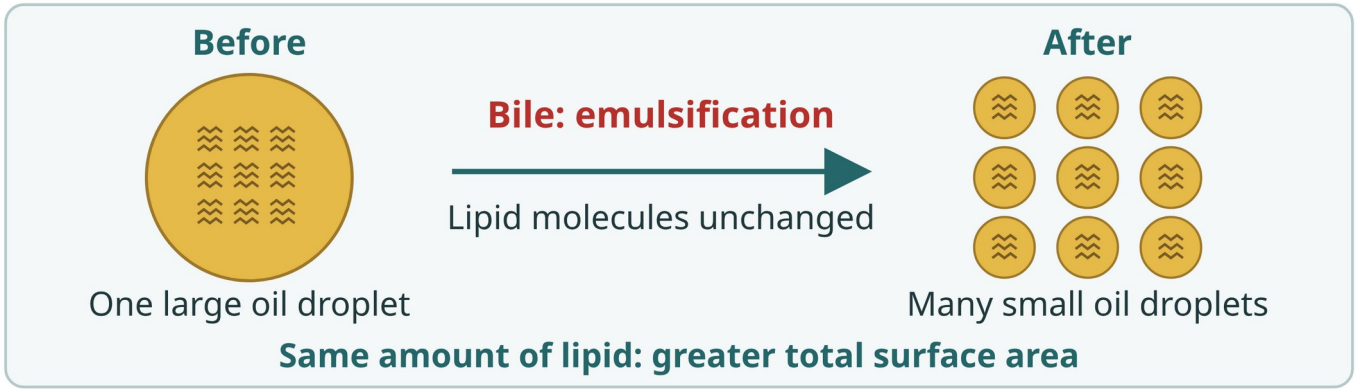
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Read the topic notes first, then apply the concepts to analyse the results.

Topic notes

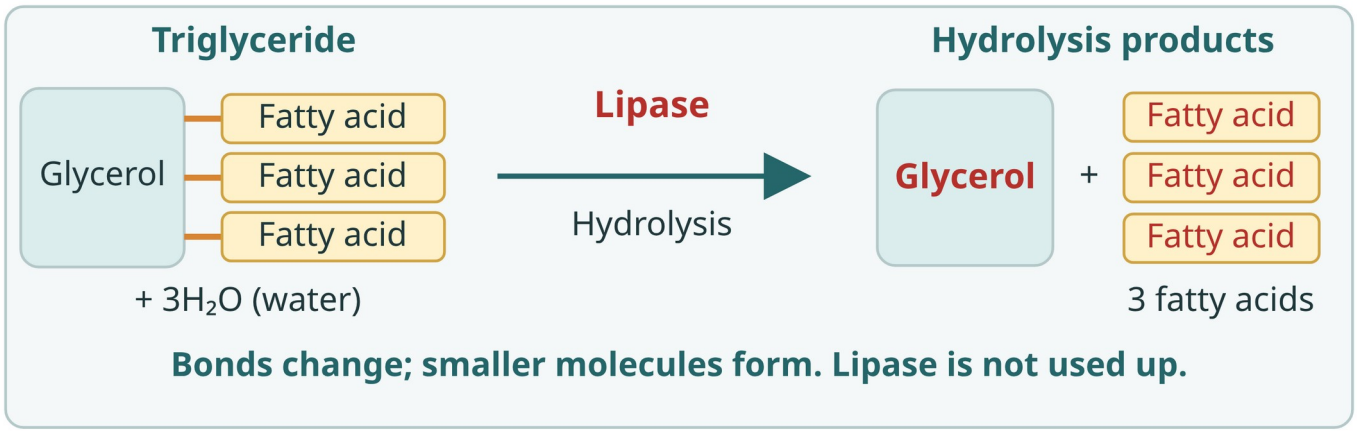
01 Bile: emulsifying lipids

In this model, X is **bile**. Made in the liver and stored in the gall bladder, **bile** contains no digestive enzymes. It **emulsifies** large oil droplets into **small droplets**, increasing the **total surface area** for **lipase** to act on. This is **physical digestion**: lipid molecules remain unchanged.



02 Lipase in pancreatic juice: chemical digestion

In this model, Y is **pancreatic juice** containing **lipase**. **Lipase** catalyses **chemical digestion** of lipids. This course uses a simplified complete-hydrolysis model: **triglycerides** form smaller **glycerol** and **fatty-acid** molecules for absorption. **Lipase** is a **protein**; high heat can **denature** it and stop it working.



Context: lipids need digestion before absorption. What happens after adding digestive juice X or Y?



Part 1 | Context and fair comparisons

Researchers added 2 mL X or 2 mL Y to identical oil–water mixtures, shook them identically at 37°C, and compared appearances after one hour.

1. Identify the variables: choose all appropriate factors in each row.

IV(the factor changed on purpose)；DV(the result you measure or observe)；CV(the factors kept the same)

Independent variable	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ Combination of digestive juices
Dependent variable	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ Combination of digestive juices
Controlled variables	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ Combination of digestive juices

2. Select the assumptions for a fair comparison (choose all that apply)

(An assumption is a condition we have not proved but take as true, to help make the experiment results valid.)

□ Observe all setups after the same reaction time.

□ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.

□ Use the same method of shaking for every setup.

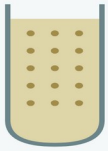
□ The X setup may receive more total liquid than the Y setup.

Part 1 | Apply concepts to the individual actions of X and Y

Both model results show one hour after adding the juice and shaking.

Set X

Add 2 mL X



Cloudy emulsion

Set Y

Add 2 mL Y



Thinner oil layer; two layers remain

Apply emulsification and surface-area concepts to analyse how X helps Y.

Researchers used two fresh tubes, added X or equal water and shook them, then added the same amount of Y. Pretreatment times were equal.

Part 2 | Extension 1: how does X help Y?

The following model results show one hour after adding Y.

X first, then Y

1 mL X → 1 mL Y



Less cloudy
No obvious oil layer

Water first, then Y

1 mL water → 1 mL Y



Thinner oil layer
Two layers remain

3. How does X help Y act?

X disperses large oil droplets into smaller droplets. The total surface area of the same amount of lipid:

□ increase

□ decrease

Part 2 | Extension 1: further analysis

4. What does appearance tell us?

□ Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol and fatty acids have formed.

□ Disappearance of the oil layer completely proves that lipids have broken down into glycerol and fatty acids.

Part 3 | Extension 2: effects of boiling

Researchers used unboiled X followed by Y as the baseline. They boiled one juice, cooled it to 37°C, then added 1 mL each of X and Y in the same order; results show one hour later.

Boiled X + unboiled Y

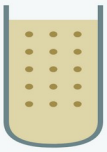
Use after cooling



Less cloudy; no oil layer

Unboiled X + boiled Y

Use after cooling



Cloudy emulsion

5. Which juice loses its action after boiling?

□ Digestive juice X

□ Digestive juice Y

Integrated explanation

6. Using the experimental results and topic notes, explain how bile and lipase in pancreatic juice work together to help digest lipids, and how boiling affects their actions.